PyGame Flappy Bird Tutorial - Part 5

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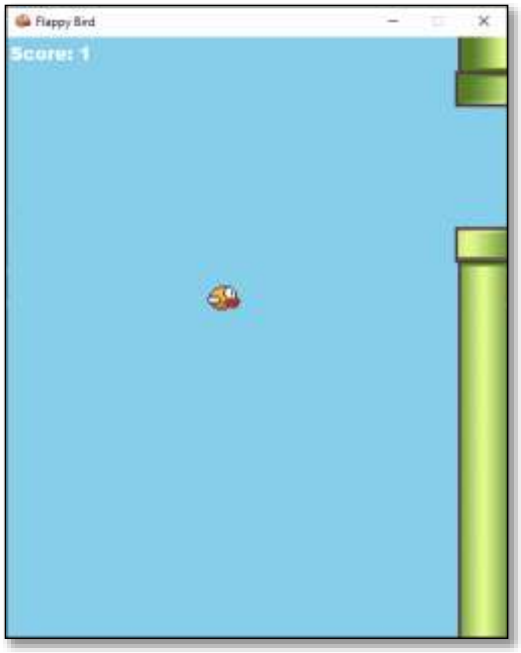
Time required: 30 minutes

Preview of the Game

Here's a sneak peak of the game that we are going to work on.

Flappy Bird Demo Video



Reset the Pipes

We only get one set of pipes. Let's reset the pipes to have more pipes traveling across.

Save **flappy_bird_4.py** as **flappy_bird_5.py**

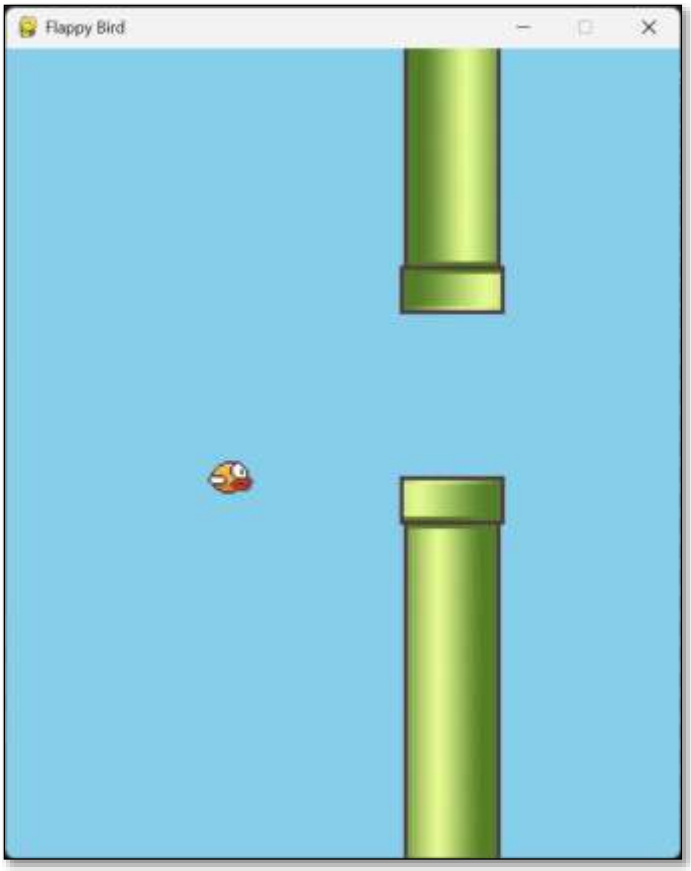
Let's add a **reset_pipes** method.

```
    # -------------------- RESET PIPES -----------------------------------#
    def reset_pipes(self):
        """Reset pipes every time they leave the screen"""
        # Pick a random height for the bottom of the top pipe
        self.pipe_upper_rect.bottom = randint(
            50,                 # Set minimum random number to 50
            config.HEIGHT // 2  # Set maximum to half the surface height
        )

        # Set lower pipe top to upper pipe bottom plus pipe gap
        self.pipe_lower_rect.top = self.pipe_upper_rect.bottom \
            + self.pipe_gap_size

        # Set initial X off screen to right
        self.pipe_upper_rect.left = config.WIDTH
        self.pipe_lower_rect.left = config.WIDTH
```

Call reset_pipes from the game loop.

```python
    # ---------------------------- GAME LOOP ------------------------------- #
    def game_loop(self):
        """Infinite game loop"""
        while True:
            self.check_events()
            self.draw_bird()

            # If the pipes are off the screen, reset them
            if self.pipe_upper_rect.right < 0:
                self.reset_pipes()

            # ----------------- MOVE PIPES ------------------------------- #
            # Move pipe images from right to left
            self.pipe_upper_rect.left = self.pipe_upper_rect.left - self.pipe_move
            self.pipe_lower_rect.left = self.pipe_lower_rect.left - self.pipe_move

            # Draw pipes to the surface
            self.surface.blit(
                self.pipe_lower,  # Source image
                self.pipe_lower_rect,  # Destination location of image
            )
            self.surface.blit(
                self.pipe_upper,  # Source image
                self.pipe_upper_rect,  # Destination location of image
            )

            # Cap game speed at 60 frames per second
            self.clock.tick(60)

            # ----------------- UPDATE DISPLAY ---------------------------- #
            # From surface, update Pygame display to reflect any changes
            pygame.display.update()

            # Cap game speed at 60 frames per second
            self.clock.tick(60)
```

Example run:



You can fly your bird up and down and through the pipes. The pipes keep coming!

There are a few issues. The bird can fall off the screen or fly up to the sun. There aren't any collisions or score keeping.

Coming right up!

Assignment Submission

1. Attach a screenshot showing the operation of the program.
2. Zip up the program files folder and submit in Blackboard.